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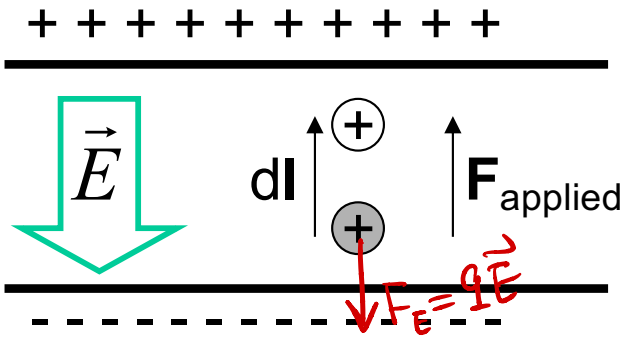
Lectures 5

Potential Functions for Static Fields

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Electric Potential



How much potential energy does a charge have in an electric field?

Answer: It depends on the work that you had to do to get it to that spot. You have to work against the E-field.

$$\Delta PE = \vec{F}_{applied} \bullet d\vec{l} = -\vec{F}_e \bullet d\vec{l}$$

$$\vec{F}_e = q\vec{E}$$

$$\Delta PE = -q\vec{E} \bullet d\vec{l}$$

Answer depends on the amount of charge q
Can we define something determined by the field only

Analogy to gravity: $\vec{F}_g = m\vec{g},\ PE = mgh,\ \dfrac{PE}{m} = gh$

Electric Potential - Definition

Work you do per unit charge is the ELECTRIC POTENTIAL DIFFERENCE between the two points

$$\Delta V = \frac{\Delta PE}{q} = -\vec{E} \bullet d\vec{l} = -E_x \Delta x \ \ if \ d\vec{l} = \Delta x \hat{a}_x$$

Units: $\frac{Nm}{C} = \frac{J}{C} = VOLTS$

$$E_x = -\frac{\Delta V}{\Delta x}, E_y = -\frac{\Delta V}{\Delta y}, E_z = -\frac{\Delta V}{\Delta z}$$ Units: $\frac{V}{m}$

$$\vec{E} = -\nabla V$$

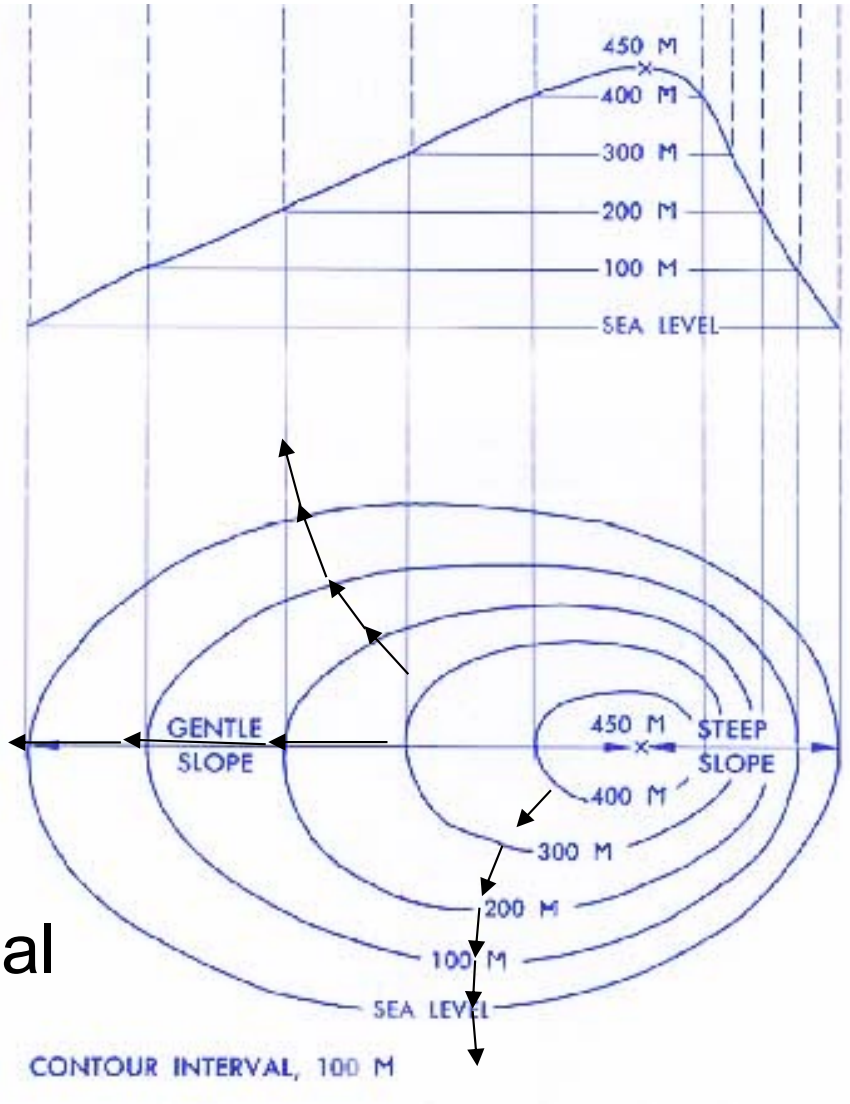
Gradient of E $\nabla V \equiv \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right)$

Potential: Mount Electron



What is the steepest slope on the mountain?

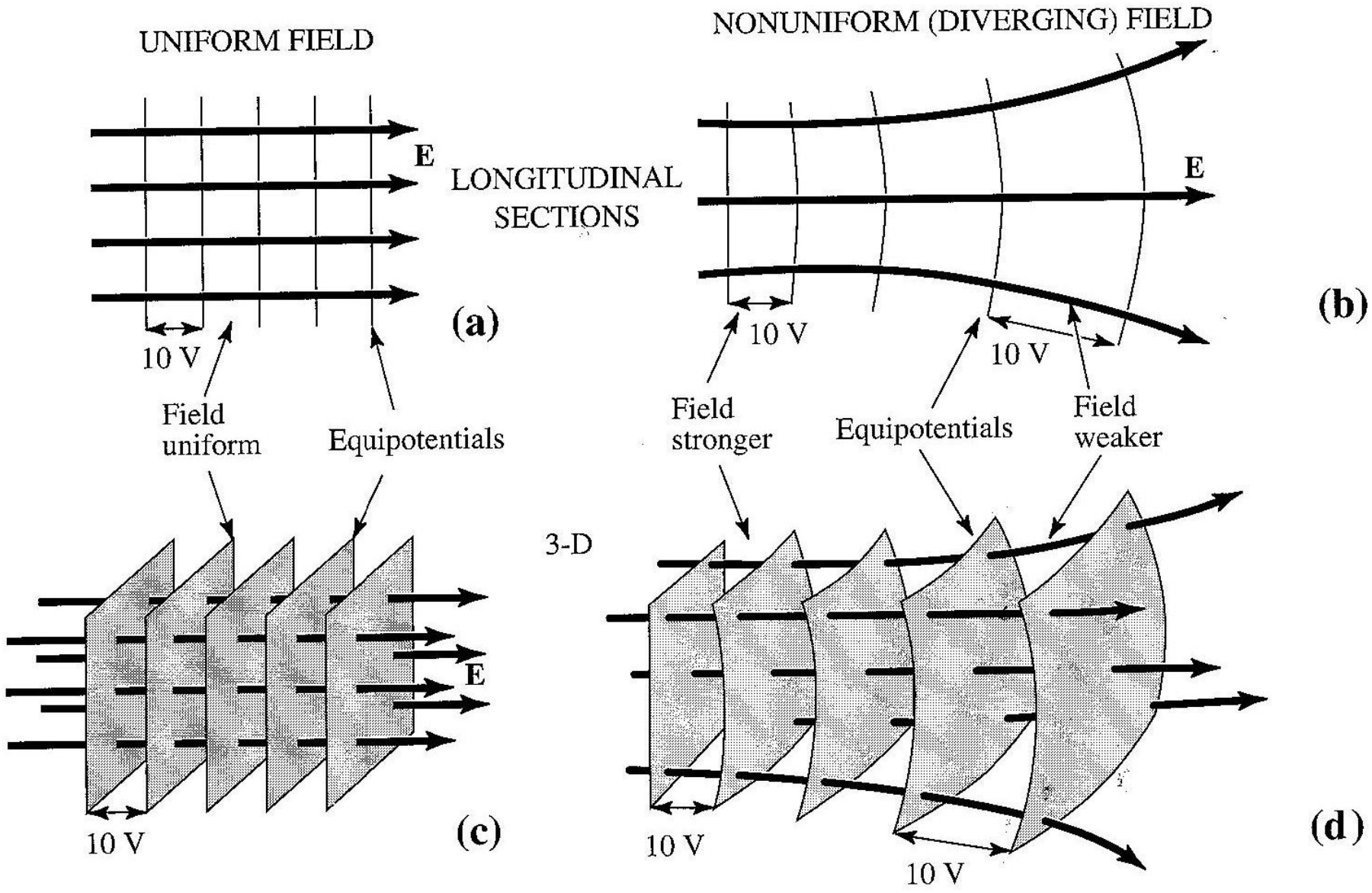
Potential



Surfaces of equal potential

What direction is the steepest slope at a given point?

Ans: The steepest downslope is ***opposite*** *the gradient! This is always perpendicular to surfaces of constant potential*


UNIFORM FIELD
NONUNIFORM (DIVERGING) FIELD
E
LONGITUDINAL SECTIONS
E
(a)
(b)
10 V
10 V
10 V
Field uniform
Equipotentials
Field stronger
Equipotentials
Field weaker
3-D
E
(c)
(d)
10 V
10 V
10 V

Gradient Definition

$$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z\right)$$

Most important definition of today:

$$\vec{E} = -\nabla V$$ V: electrostatic potential

$$\nabla = \hat{a}_x \frac{\partial}{\partial x} + \hat{a}_y \frac{\partial}{\partial y} + \hat{a}_z \frac{\partial}{\partial z}$$

Or $$\nabla V = \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right)$$

Del **Operator** (Cartesian Coordinates)
used to do gradient, divergence, and curl

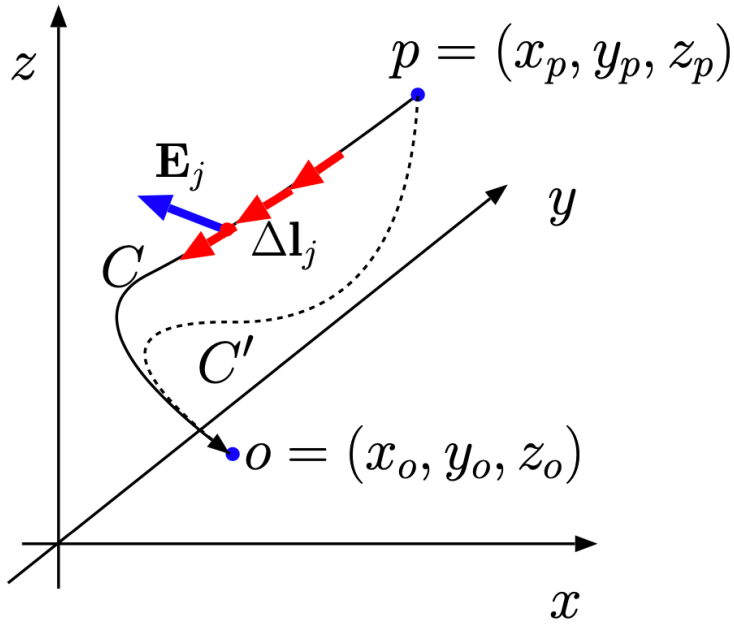
Electrostatic Potential V and electric field E

From V to E

$$\vec{E} = -\nabla V$$

From E to V

$$V = -\int_o^P \vec{E} \cdot d\vec{l} = \int_P^o \vec{E} \cdot d\vec{l}$$



$$d\mathbf{l} = \hat{x}dx + \hat{y}dy + \hat{z}dz = (dx, dy, dz)$$

$$\mathbf{E} \cdot d\mathbf{l} = (E_x, E_y, E_z) \cdot (dx, dy, dz) = E_x dx + E_y dy + E_z dz.$$

$$\int_p^o \mathbf{E} \cdot d\mathbf{l} = \int_p^o (E_x dx + E_y dy + E_z dz)$$

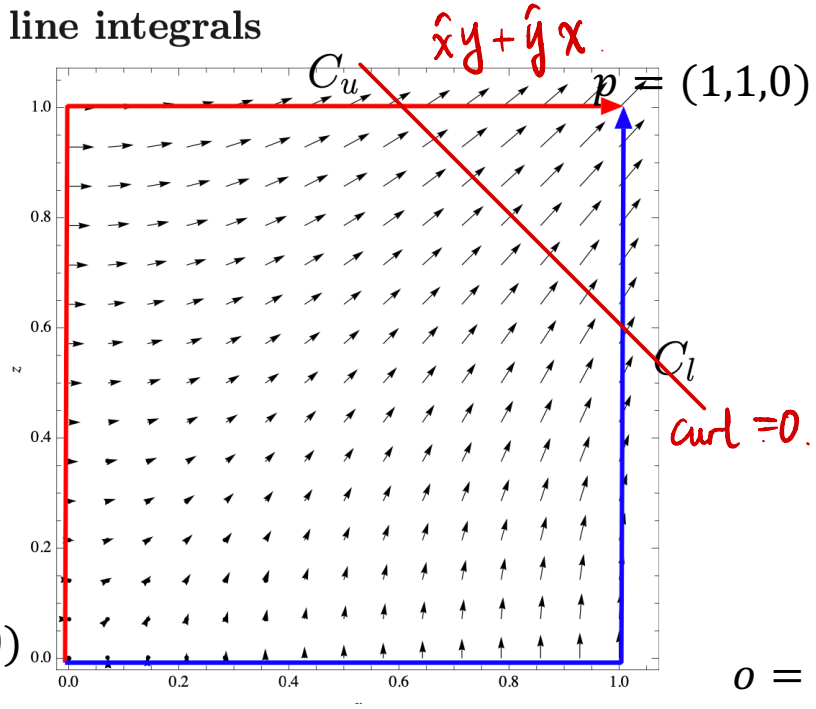
integration will in general be path dependent except for when E is curl-free

Example: Path Independent and Curl Free

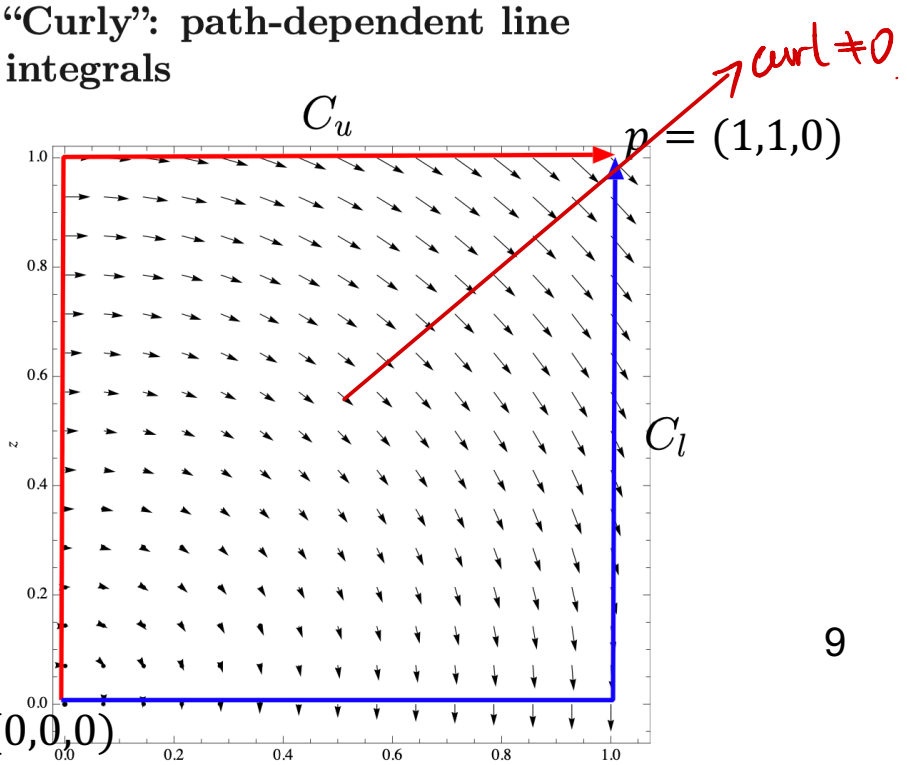
The field $\vec{\boldsymbol{E}} = \hat{x}y \pm \hat{y}x$ is curl-free with the + sign, but not with − sign.
Calculate the line integral of $\vec{\boldsymbol{E}}$ (for both signs, ±) from a point $o = (0,0,0)$ to point $p = (1,1,0)$ for two different paths.

$$\nabla \times (\hat{x}y \pm \hat{y}x) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & \pm x & 0 \end{vmatrix} = \hat{z}(\pm 1 - 1)$$

Curl-free: path-independent line integrals



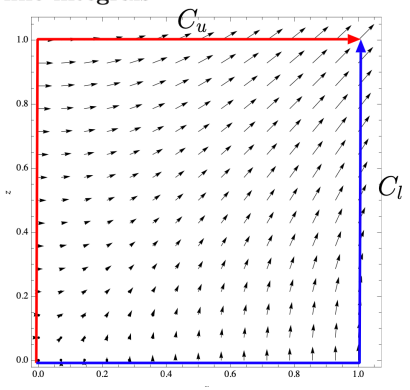
"Curly": path-dependent line integrals



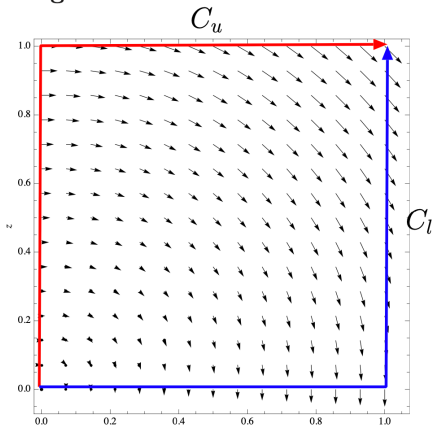
Path Independent and Curl Free (cont.)

The field $\vec{E} = \hat{x}y \pm \hat{y}x$ is curl-free with the + sign, but not with − sign.
Calculate the line integral of $\vec{E}$ (for both signs, ±) from a point $o = (0,0,0)$ to point $p = (1,1,0)$ for two different paths.

Curl-free: path-independent line integrals



"Curly": path-dependent line integrals



$$\int_o^p \mathbf{E}\cdot d\mathbf{l} = \int_o^p (E_x dx + E_y dy + E_z dz) = \int_o^p (y\,dx \pm x\,dy)$$

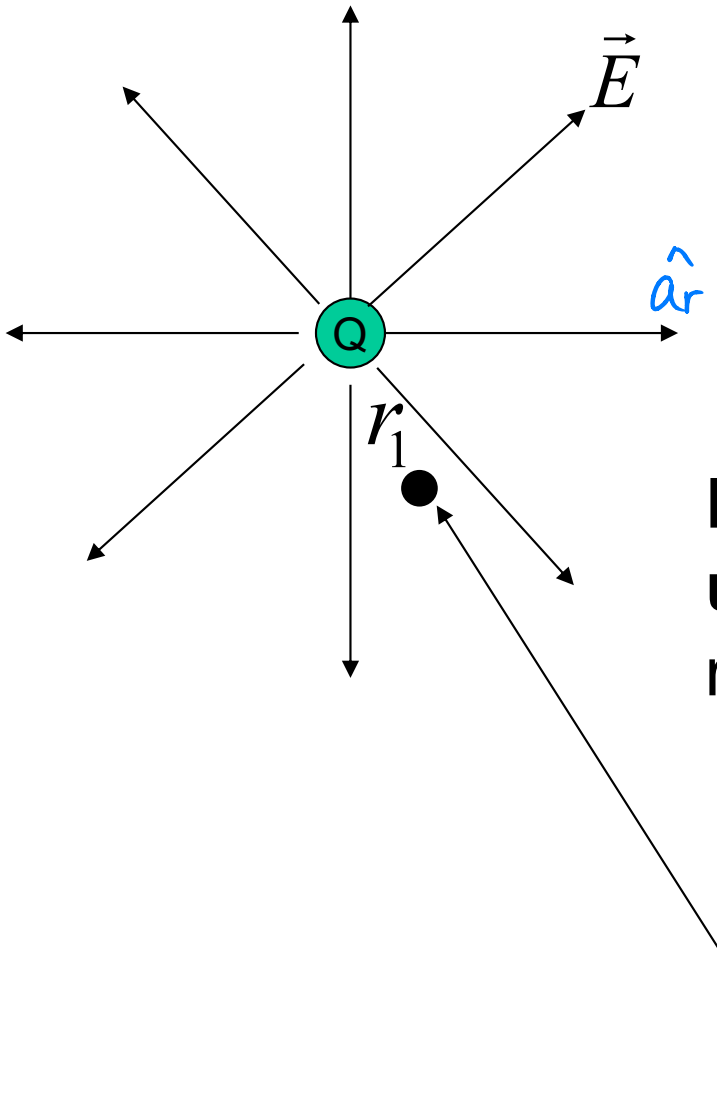
C_u path. $\int_o^p \vec{E}\cdot d\vec{\ell} = \int_{y=0}^{1} y\,dx \pm x\,dy \Big|_{x=0} + \int_{x=0}^{1} y\,dx \pm x\,dy \Big|_{y=1}$

$= 0 + \int_{x=0}^{1} 1\cdot dx = 1.$

C_ℓ path $\int_o^p \vec{E}\cdot d\vec{\ell} = \int_{x=0}^{1} y\,dx \pm x\,dy \Big|_{y=0} + \int_{y=0}^{1} y\,dx \pm x\,dy \Big|_{x=1}$

$= 0 + \int_{y=0}^{1} \pm dy = \pm 1.$

Example: Potentials for a Point Charge



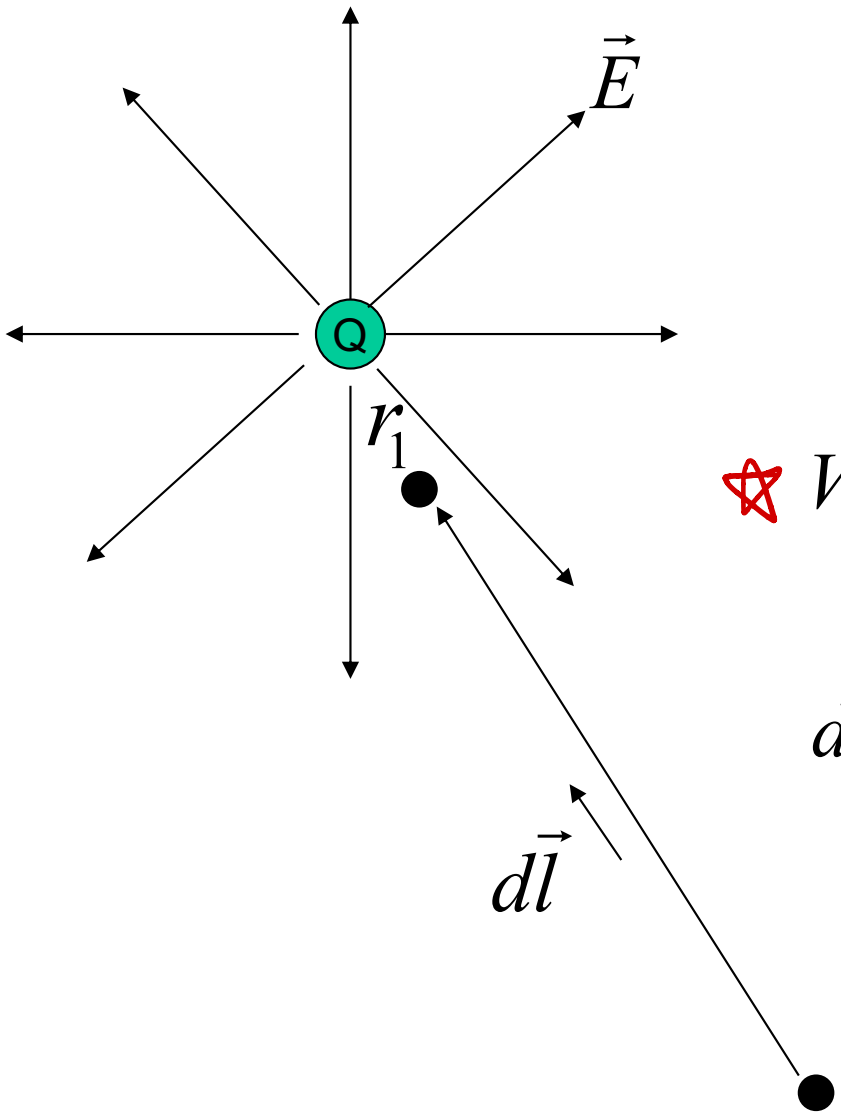
$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{a}_r$$

In spherical coords

How much work is required to move a **unit** charge from infinity to a radial distance = r_1

at ∞, $V_\infty = 0$.

Example: Potentials for a Point Charge



$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{a}_r$$

In spherical coords

$$\vec{E} = -\nabla V$$

$$V(r) = -\int_{\infty}^{r_1} \vec{E} \bullet d\vec{l}$$

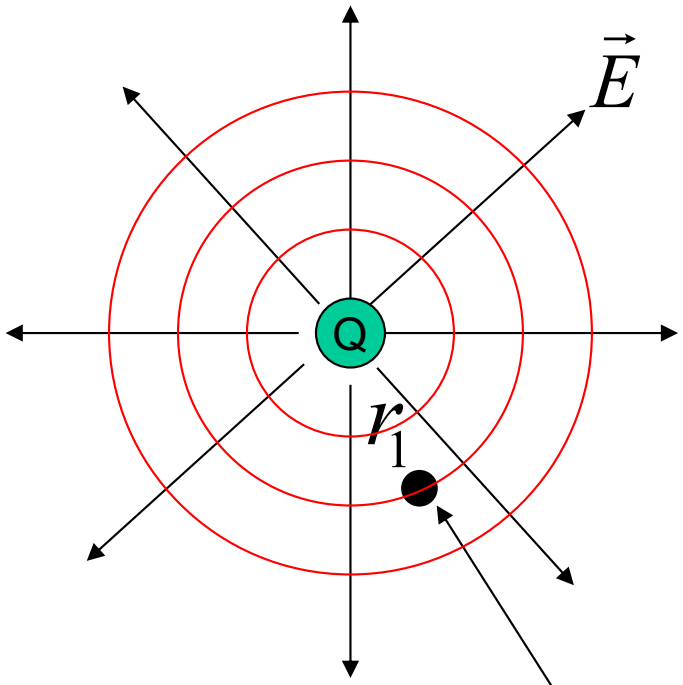
"Absolute" potential at r_1 using zero potential at $r = \infty$

$$d\vec{l} = -|dr|\hat{a}_r = dr\hat{a}_r$$

Since dr<0 going from r=∞

$$V(r) = -\int_{\infty}^{r_1} \frac{Q}{4\pi\varepsilon_0 r^2} dr$$

Example: Potentials for a Point Charge



$$V(r) = -\int_{\infty}^{r_1} \frac{Q}{4\pi\varepsilon_0 r^2}\, dr$$

$$= \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{\infty}\right)$$

$$\boxed{= \frac{Q}{4\pi\varepsilon_0 r_1}}$$

[V].

Surfaces of constant potential are spheres in 3D
-same amount of work

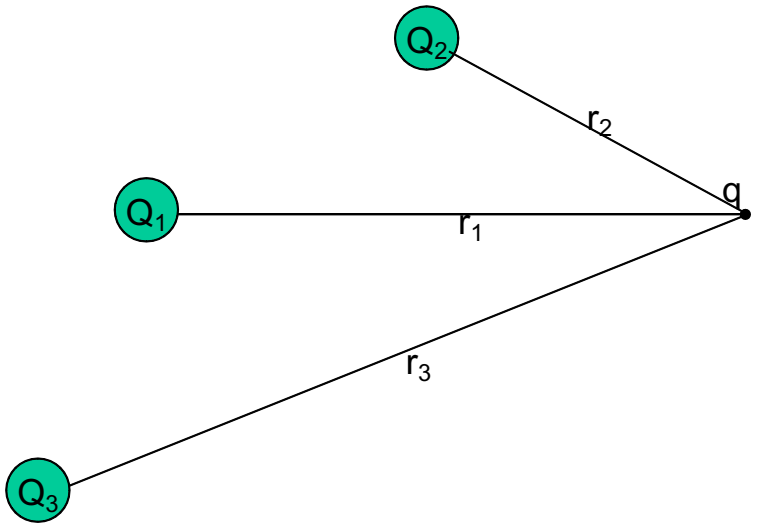
Potential Superposition

Potentials of more than one point charge are superimposed by addition

$$V_{\text{point}} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q_1}{r_1} + \frac{Q_2}{r_2} + \frac{Q_3}{r_3} + \bullet\bullet\bullet\right)$$

scalar.

[V]



Scalar Potentials

- For: $\vec{E} = yz\hat{a}_x + (y + zx)\hat{a}_y + xy\hat{a}_z$

(a)Find the scalar potential if V(0,0,0)=0. (Hint: Use a direct line path.

(b)Using 3 separate segments along x, y, and z directions.)

a) $V(x,y,z) = -\int_{(0,0,0)}^{(x,y,z)} (yz\hat{x} + (y+zx)\hat{y} + xy\hat{z}) \cdot (dx\hat{x} + dy\hat{y} + dz\hat{z})$

$= -\int_{(0,0,0)}^{(x,y,z)} yz\,dx + (y+zx)\,dy + xy\,dz$

$= -\int_{(0,0,0)}^{(x,y,z)} [d(xyz) + y\,dy]$

$= -xyz - \frac{1}{2}y^2$

Adapted from D 5.4 (p299) of old book

$\vec{E} = -\nabla V$

$V = -\int \vec{E} \cdot d\vec{l}$

Scalar Potentials

- For: $\vec{E} = yz\hat{a}_x + (y + zx)\hat{a}_y + xy\hat{a}_z$

(a)Find the scalar potential if V(0,0,0)=0. (Hint: Use a direct line path.

(b)Using 3 separate segments along x, y, and z directions.)

b). $V(x,y,z) = -\int_{x=0}^{x} yz\,dx + (y+zx)\,dy + xy\,dz \Big|_{y=0,z=0} \to 0$

$-\int_{y=0}^{y} yz\,dx + (y+zx)\,dy + xy\,dz \Big|_{x=x,z=0} \to -\int_{y=0}^{y} y\,dy$

$-\int_{z=0}^{z} yz\,dx + (y+zx)\,dy + xy\,dz \Big|_{x=x,y=y} \to -\int_{z=0}^{z} xy\,dz$

$= 0 - \frac{1}{2}y^2 - xyz$

Adapted from D 5.4 (p299) of old book

Gradient Function in Cylindrical and Spherical Coordinates

Φ = scalar field (like voltage)

$$\nabla\Phi = \begin{cases} \left(\frac{\partial\Phi}{\partial x}\hat{a}_x + \frac{\partial\Phi}{\partial y}\hat{a}_y + \frac{\partial\Phi}{\partial z}\hat{a}_z\right) & \text{Cartesian} \\ \left(\frac{\partial\Phi}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial\Phi}{\partial\phi}\hat{a}_\phi + \frac{\partial\Phi}{\partial z}\hat{a}_z\right) & \text{Cylindrical} \\ \left(\frac{\partial\Phi}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial\Phi}{\partial\theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial\Phi}{\partial\phi}\hat{a}_\phi\right) & \text{Spherical} \end{cases}$$

Lecture 5 Summary

- Since $\nabla \cdot \vec{\boldsymbol{D}} = \rho$ and $\vec{\boldsymbol{D}} = \epsilon_o \vec{\boldsymbol{E}}$, use vector identity $\nabla \cdot \nabla \Phi = 0$

$$\vec{E} = -\nabla \Phi$$

satisfy Gauss' Laws and is valid for static fields